IMAGE ENHANCEMENT

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('image.jpg')

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()


gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")
plt.show()


equalized = cv2.equalizeHist(gray)


plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(equalized, cmap='gray')
plt.title("Histogram Equalized")
plt.axis("off")

plt.show()


plt.hist(gray.ravel(), bins=256)
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()


plt.hist(equalized.ravel(), bins=256)
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```

```
kernel = np.ones((5,5), np.uint8)


erosion = cv2.erode(gray, kernel, iterations=1)

plt.imshow(erosion, cmap='gray')
plt.title("Erosion")
plt.axis("off")
plt.show()


dilation = cv2.dilate(gray, kernel, iterations=1)

plt.imshow(dilation, cmap='gray')
plt.title("Dilation")
plt.axis("off")
plt.show()


opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)

plt.imshow(opening, cmap='gray')
plt.title("Opening")
plt.axis("off")
plt.show()


closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)

plt.imshow(closing, cmap='gray')
plt.title("Closing")
plt.axis("off")
plt.show()


plt.figure(figsize=(12,8))

plt.subplot(2,3,1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(erosion, cmap='gray')
plt.title("Erosion")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(dilation, cmap='gray')
plt.title("Dilation")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(opening, cmap='gray')
plt.title("Opening")
```

```
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(closing, cmap='gray')
plt.title("Closing")
plt.axis("off")

plt.tight_layout()
plt.show()
```